

AtlasJawSpeech000003

Therapeutic Protocols: Control and Finesse in Orofacial Rehabilitation

8.0 Introduction to Phase 2: The Transition from Stability to Finesse

The rehabilitation of the orofacial complex is a disciplined architectural process, moving from the foundational bedrock of gross motor stability to the intricate spires of articulatory finesse. While Phase 1 of therapeutic intervention typically prioritizes the establishment of a static "anchor"—building raw strength and reducing gross instability—Phase 2, designated as "Jaw Grading (The Finesse)," represents a sophisticated evolution in neuromuscular control. This phase is not merely about strength; it is about the *differentiation* of movement, the ability to arrest the mandible at precise, fractional apertures without reliance on the primitive "open-shut" binary that characterizes immature or disordered oral motor patterns.¹

Jaw grading is clinically defined as the controlled segmentation of mandibular movement through various vertical heights—specifically high, medium, and low excursions—executed with fluid dissociation from the tongue and lips.³ It serves as the physiological bridge between static structural integrity and the dynamic requirements of articulate speech and rotary mastication. Without adequate grading, the mandible often defaults to a "fixing" pattern, where the speaker or eater recruits extraneous musculature from the neck, shoulders, or lips to stabilize the jaw, leading to reduced intelligibility, vowel distortion, and inefficient bolus management.⁴

This comprehensive research report serves as an exhaustive clinical manual for "8.2 Phase 2: Jaw Grading (The Finesse)." It integrates data from Oral Placement Therapy (OPT), myofunctional sciences, and neurodevelopmental reflex integration to provide a robust framework of exercises, methodologies, and theoretical underpinnings necessary to achieve true mandibular control. The following sections dissect the therapeutic protocols required to decouple the jaw from the body, establish mechanical grading through bite blocks, and translate that stability into functional speech and feeding skills.

8.1 The Biomechanics of "Finesse" and Mandibular Control

To understand the necessity of Phase 2, one must first appreciate the unique biomechanics of the temporomandibular joint (TMJ). Unlike the "saddle" or "hinge" joints of the skeletal system that derive significant stability from their geometric congruency, the TMJ relies heavily on a complex suspension of musculature for its positional accuracy. "Finesse," in this context, refers to the neuromuscular ability to sustain the mandible at mid-range positions—points of

instability where the elevators (masseter, temporalis, medial pterygoid) and depressors (digastric, mylohyoid, geniohyoid) must engage in a delicate co-contraction to counteract gravity and tension.⁶

Clinical observations indicate that clients with deficits in jaw grading often display a "sliding" or "jutting" motion, translating the mandible laterally or anteriorly when attempting to open wide.⁷ This compensatory movement attempts to find stability by locking the condyle against the eminence, rather than relying on muscular control. Phase 2 interventions are designed to strip away these compensations, forcing the proprioceptive system to learn the internal sensation of "High," "Medium," and "Low" jaw postures without visual mirroring or skeletal locking.⁹

- **High Jaw Position:** This requires a sustained, isometric contraction of the elevators with minimal recruitment of the depressors. It correlates biomechanically with the production of high vowels such as /i/ (as in "see") and /u/ (as in "moon").⁵
- **Medium Jaw Position:** This represents a neutral, "floating" state where elevators and depressors maintain a balanced tension. It is essential for mid-vowels like /e/ (as in "bed") and /o/ (as in "go") and is often the most difficult position for clients to maintain due to the lack of end-range feedback.⁶
- **Low Jaw Position:** This necessitates the controlled lengthening (eccentric contraction) of the elevators and active contraction of the depressors. It correlates with low vowels like /a/ (as in "hot") and /æ/ (as in "cat") and requires the greatest degree of excursion.⁶

8.2 The "Body Connection": Reflex Integration

Prerequisites

The title "Jaw, Teeth, Speech, and Body Connection" implies a profound physiological truth: distal motor function influences proximal stability. Before a client can successfully engage in direct jaw grading exercises, the clinician must ensure that the primitive reflexes connecting the hand, mouth, and body are sufficiently integrated. If these reflexes remain active, they act as neurological "noise," coupling the jaw to the hands or torso and preventing the isolated, dissociated movements required for grading.¹²

8.2.1 The Babkin Palmomental Reflex: The Hand-Mouth Link

The Babkin Palmomental reflex is perhaps the most significant inhibitor of jaw grading finesse. Physiologically, this reflex links stimulation of the palm (specifically the thenar eminence) to the involuntary opening of the mouth or movement of the jaw.¹⁴ In infants, this facilitates breastfeeding—kneading the breast triggers mouth opening. However, in a developing child or adult rehabilitation client, a retained Babkin reflex creates a "synkinetic" interference. When the client attempts a fine motor task with their hands (like writing or using a tool), their jaw may involuntarily open or clench. Conversely, when they attempt to grade their jaw for speech,

their hands may curl or twitch.¹⁶

Clinical Implication: A client with a retained Babkin reflex cannot achieve true jaw "finesse" because the neural pathway couples the jaw to the hands. Any therapeutic attempt to grade the jaw without addressing this reflex will likely result in frustration or compensatory recruitment of the hands to "help" the mouth.¹⁸

Protocol A: Reflex Integration Exercises

To prepare the system for Phase 2 Jaw Grading, the following integration exercises should be performed. These activities are designed to desensitize the reflex arc and separate hand movement from oral motor output.

Exercise 1: The Babkin Integration Massage

This technique aims to desensitize the specific trigger zones on the hands that elicit the oral response.¹⁴

1. **Positioning:** The client should be seated or supine in a relaxed state. The clinician or caregiver holds the client's hand, exposing the palm.
2. **Stimulation:** Using a firm, deep pressure (light touch can be over-stimulating), the clinician massages the thenar eminence (the fleshy pad at the base of the thumb).
3. **Active Observation:** While massaging, the clinician must watch the client's mouth intently. Look for micro-movements: a twitch of the lip, a slight parting of the teeth, or a tightening of the mentalis (chin) muscle.
4. **Inhibition Cue:** If oral movement is detected, instruct the client: "Keep your lips closed and your jaw relaxed." The goal is for the client to *feel* the pressure on the hand without *reacting* with the mouth.
5. **Duration:** Perform for 30-60 seconds on each hand.
6. **Progression:** As the client tolerates the massage without oral reaction, introduce *active* finger opposition. Ask the client to touch their thumb to their pinky finger repeatedly while maintaining a relaxed, closed mouth. This moves from passive desensitization to active cortical override of the reflex.²⁰

Exercise 2: Isometric Hand-Jaw Dissociation ("The Meatball")

This exercise, often referred to in neurodevelopmental protocols as "The Meatball," targets the core stability required to support the head and neck, thereby freeing the jaw from postural holding duties.²²

1. **Starting Position:** The client lies on their back (supine) on a firm surface.
2. **Action:** The client curls their body inward, bringing their knees toward their chest and lifting their head off the floor, crossing their arms over their chest. They should resemble a "ball" or "meatball."
3. **The Hold:** Maintain this flexed posture for 10 to 20 seconds.
4. **The Constraint (Crucial for Jaw Grading):** During the intense core exertion, the client must maintain a *relaxed jaw*. The teeth should be slightly apart, and the lips gently closed. They must breathe rhythmically.

5. **Clinical Rationale:** Often, clients will clench their teeth or jut their jaw to help "hold" the head up. By forcing the core (rectus abdominis) and neck flexors (sternocleidomastoid) to do the work while the jaw stays loose, we are breaking the "fixing" pattern that inhibits grading.
6. **Modification:** If the client cannot lift their head, they can keep the head on the floor and focus solely on the limb flexion while keeping the jaw relaxed.²²

Exercise 3: The Zombie Walk (ATNR/STNR Integration)

While primarily targeting the Asymmetrical Tonic Neck Reflex (ATNR), this exercise supports jaw grading by dissociating head rotation from limb extension, a prerequisite for independent jaw movement.²²

1. **Position:** Client stands with arms extended straight out in front (like a zombie), fingers splayed.
2. **Action:** The client walks forward 10 steps.
3. **Rotation:** While walking, the client turns their head fully to the left, then fully to the right.
4. **Constraint:** The arms must remain pointing straight forward. They should not drift in the direction of the head turn. The jaw must remain relaxed; no clenching with the head turn.
5. **Relevance to Phase 2:** This creates a stable "body platform" where the head can move independently of the arms, and by extension, the jaw can move independently of the head.²³

8.3 Phase 2: Mechanical Jaw Grading Protocols (Bite Blocks)

Once the neurological foundation is stabilized via reflex integration, therapy proceeds to **Mechanical Grading**. This sub-phase utilizes external tools—specifically graduated bite blocks—to provide the tactile and proprioceptive feedback necessary for the client to locate and sustain specific jaw heights. The "TalkTools® Jaw Grading Bite Blocks" or "Ark Therapeutic® Bite Blocks" are the industry standards for this protocol.⁷

The grading hierarchy is generally divided into sets of blocks, numbered #2 through #7 (TalkTools) or graded by thickness (Ark), corresponding to the physiological apertures required for speech and feeding.

Block Level	Jaw Position	Vowel Correlation (Phonetic Target)	Muscle Action Focus
Block #2	High	/i/ (ee), /u/ (oo)	Elevator Dominance (Shortened

			Masseter)
Block #3	High-Mid	/I/ (ih)	Fine Elevator Control
Block #4	Mid	/e/ (ay), /E/ (eh)	Co-contraction (Elevators/Depressors Balanced)
Block #5	Low-Mid	/o/ (oh)	Depressor Activation
Block #6	Low	/a/ (ah)	Depressor Dominance / Elevator Lengthening
Block #7	Lowest	/æ/ (cat)	Maximum Excursion

8.3.1 Protocol B: Isometric Bite and Hold (The Foundation)

The primary objective of this protocol is to build "holding power" at each grade. It is insufficient to simply open the mouth to the correct height; the client must be able to *sustain* that height against resistance and time, creating a stable anchor for the tongue.²⁴

Detailed Methodology:

1. **Selection:** Begin with **Block #2 (High)**. This is biomechanically the most stable position as the condyle is seated in the fossa, requiring less muscular stabilization than the open positions.²⁶
2. **Placement:** Place the block laterally between the pre-molars or molars on the *right* side. The block should be parallel to the teeth. Avoid placing it on the incisors for this exercise, as the goal is posterior stability.²⁴
3. **Instruction:** Give the verbal cue, "Bite and hold."
4. **Duration:** The client must sustain a firm bite for 15 seconds.
5. **Observation & Troubleshooting:**
 - *Sliding:* If the jaw slides toward the block, it indicates weakness on the contralateral side. Correction: Use a mirror to provide visual feedback on symmetry.⁹
 - *Lip Pursing:* If the lips purse or tighten, the client is using the orbicularis oris to assist the jaw. Correction: Gently pull the lower lip down or tap the chin to disengage the mentalis.²⁷

6. **Symmetry:** Repeat the exact procedure on the *left* side.
7. **Progression:** Once the client can hold Block #2 for 15 seconds bilaterally with no compensation, proceed to Block #3, then #4, and so on, down to Block #7. Note that the lower blocks (#5, #6, #7) are significantly more challenging for clients with low tone, as the muscles are stretched and mechanically disadvantaged.¹

8.3.2 Protocol C: Tug and Pull (Stability Under Tension)

Static holding is a prerequisite, but functional activities like chewing and speaking involve dynamic forces. The "Tug and Pull" exercise introduces destabilizing forces that the client must resist, simulating the varying pressures of a bolus.²⁴

Detailed Methodology:

1. **Placement:** Insert the appropriate block (start with #2 or #3) between the molars.
2. **Action:** Instruct the client to "Bite and don't let me pull it out."
3. **Force Application:** The clinician applies a gentle, rhythmic pulling force on the stick—pulling anteriorly (out of the mouth) for 1 second, then relaxing for 1 second, without the client letting go.
4. **Duration:** Perform 10 "tugs" per side.
5. **Clinical Note:** Watch the client's head. If the head jerks forward with each tug, they are using their neck muscles (sternocleidomastoid) to stabilize rather than their jaw elevators. Cue: "Keep your head still, hold with your teeth".²⁴

8.3.3 Protocol D: Alternating Grading (The Switch)

This exercise trains the *transition* between grades, which is the mechanical essence of co-articulation in speech (e.g., moving from "tea" to "top"). It targets the agility of the system.²⁵

Detailed Methodology:

1. **Setup:** Have two blocks ready: one High (e.g., Block #2) and one Low (e.g., Block #6).
 2. **Execution:**
 - Place Block #2 (Right). "Bite." (Hold 2 seconds). Remove.
 - Immediately place Block #6 (Right). "Bite." (Hold 2 seconds). Remove.
 - Repeat this alternating sequence 10 times rapidly.
 3. **Goal:** The client should be able to drop the jaw to the exact height of Block #6 and close to Block #2 without "searching," fumbling, or sliding laterally. The movement should be a clean vertical piston action.²
 4. **Advanced Variation:** Use Block #3 and Block #5 (Mid-range transitions). This is often harder because the sensory difference between the two is smaller, requiring finer proprioceptive discrimination.
-

8.4 The Chewing Hierarchy: Functional Grading

While bite blocks provide isometric stability, functional grading is best developed through mastication. Lori Overland's "Chewing Hierarchy" is a widely recognized protocol within Phase 2 that systematically builds the rotary jaw movements required for managing food textures.²⁹ This hierarchy moves from simple vertical movements to complex rotary transfers.

8.4.1 Level 1: The Munch Chew (Vertical Grading)

Before a client can master the "finesse" of rotary chewing, they must master the vertical "munch." This level focuses on establishing a rhythmic, graded bite-and-release pattern.

- **Tool:** Yellow Chewy Tube (T-shaped) or a similar firm, non-food tool.
- **Placement:** Lateral molar ridge (e.g., lower first molar).
- **Method:**
 1. Present the stem of the T-tube to the molar.
 2. Instruct: "Bite, release, bite, release."
 3. **Rhythm:** Aim for a steady tempo of approximately 1 bite per second. A metronome or rhythmic tapping can assist clients with dyspraxia.³¹
 4. **Volume:** Goal is 20 consecutive bites on the right, then 20 on the left.
 5. **Success Criteria:** No sliding of the jaw. The bite should compress the tube significantly but not necessarily bottom out if the tube is very hard. The key is *rhythm* and *verticality*.²⁹

8.4.2 Level 2: Lateral Weight Transfer (The Crossing)

This level introduces the concept of moving the jaw across the midline, a critical component of rotary chewing. It requires the jaw to grade slightly open, move laterally, and then close.

- **Tool:** Two Chewy Tubes (or one moved rapidly).
- **Method:**
 1. Place Tube A on the Right Molar. "Bite."
 2. Place Tube B on the Left Molar. "Bite."
 3. **Sequence:** Alternate bites—Right, Left, Right, Left.
 4. **Focus:** The clinician observes the transition. Does the jaw drop comfortably to clear the teeth before moving to the other side? Or does the client drag the teeth across? The "Finesse" is in the clearance (opening) during the transfer.³⁰
 5. **Repetitions:** 4-5 sets of alternating bites.

8.4.3 Level 3 & 4: Rotary Movement and Bolus Management

Levels 3 and 4 introduce the grinding motion essential for breaking down fibrous foods.

- **Level 3:** Using a single tube, the client bites, and the clinician gently guides the jaw in a small circular motion to simulate the "grind."
- **Level 4 (Food Introduction):** Introduce "safe" solids like Veggie Stix or graham crackers.

Place the food on the molar. The client must bite, transfer the food with the tongue to the other side (tongue lateralization), and bite again. This functional task integrates the jaw grading stability learned in levels 1-2 with tongue mobility.²⁹

8.5 Speech Applications: The "Speech Block" Protocol

The ultimate goal of Phase 2 is to translate mechanical stability into intelligible speech. The "Speech Block" protocol uses the bite blocks as tactile cues to map the Vowel Quadrilateral onto the client's motor memory.²⁵

8.5.1 The Physiology of Vowel Grading

Standard phonetic theory categorizes vowels by tongue position (high/low, front/back). In the context of rehabilitative protocols like OPT, we emphasize the role of the *jaw* in supporting these tongue positions.

- **Problem:** A client with low tone or apraxia often has a "collapsed" vowel space. They may produce /i/ (see) and /a/ (hot) with the same neutral jaw height, relying entirely on the tongue. This results in acoustic distortion (e.g., "beat" sounding like "bit" or "bet").⁵
- **Solution:** By mechanically fixing the jaw height with a block, we force the tongue to work independently. This is known as **Jaw-Tongue Dissociation**.³⁴

8.5.2 Phase 2a: Tactile Vowel Drills

These exercises use the block to "teach" the muscles where to hold the jaw for specific sounds.

High Vowel Drill (/i/ - "ee"):

1. **Insert Block #2 (High)** between the molars.
2. **Instruction:** "Bite and say /i/."
3. **Rationale:** The block prevents the jaw from closing too tight (clenching) or dropping too low. The tongue is forced to elevate to the palate to create the constriction for /i/.
4. **Repetition:** 10 times. "Eeee... Eeee... Eeee...".²⁵

Mid Vowel Drill (/e/ - "ay"):

1. **Insert Block #4 (Mid)**.
2. **Instruction:** "Bite and say /e/ (as in 'day')."
3. **Rationale:** This trains the difficult mid-range position. The client feels the "space" required for the vowel.

Low Vowel Drill (/a/ - "ah"):

1. **Insert Block #6 (Low)**.
2. **Instruction:** "Bite and say /a/."

3. **Clinical Note:** This is often the most revealing drill. Clients with tight jaws or spasticity will struggle to keep the jaw down on the block while vocalizing. They may try to spit the block out to close the mouth. The block enforces the excursion.²⁵

8.5.3 Phase 2b: Word Lists and Transition Drills

Once isolated vowels are established with blocks, the clinician moves to real words, using the blocks only as cues if the client fails to achieve the grade. The following word lists are curated to challenge specific grading transitions.

Table 1: High Jaw Word List (Requires Block #2 Stability)

Focus: Sustained contraction of elevators.

Target Vowel	Word Examples	Clinical Cue
/i/ (High Front)	See, Bee, Tea, Key, Knee, Fee, We	"Smile slightly, keep teeth nearly touching."
/u/ (High Back)	Moon, Boot, Tool, Cool, Shoe, Food, Woo	"Lips round, jaw stays high."
/I/ (High-Mid)	Pin, Bin, Tin, Kin, Min, Fin, Win	"Jaw drops just a tiny bit from /i/."
Combined	"See the Moon", "Two Feet", "New Shoes"	Maintain high position across phrase.

Table 2: Low Jaw Word List (Requires Block #6 Excursion)

Focus: Controlled depression and relaxed elevators.

Target Vowel	Word Examples	Clinical Cue
/a/ (Low Back)	Hot, Pot, Lot, Not, Got, Shop, Top	"Drop the jaw, tongue flat."
/æ/ (Low Front)	Cat, Bat, Mat, Fat, Sat, Map, Tap	"Jaw low, tongue forward."

/au/ (Diphthong)	Cow, Pow, Wow, Now, How, Loud, Town	"Start low, glide up."
Combined	"Hot Pot", "Fat Cat", "Not Now"	Ensure full excursion on every vowel.

Table 3: Grading Transition Drills (The "Weightlifting" of Speech)

These pairs force the neuromuscular system to switch rapidly between elevator and depressor dominance. This is the core of "Finesse."

Transition Type	Word Pair Examples	Motor Challenge
High -> Low	See -> Saw Feet -> Fat Heat -> Hot Keep -> Cop	Rapid release of tension. Gravity assists, but control prevents "crashing."
Low -> High	Hot -> Heat Pop -> Peep Cop -> Keep Mop -> Me	Rapid contraction against gravity. Requires explosive power and precise braking.
Mid -> Low	Bed -> Bad Pen -> Pan Mess -> Mass	Fine discrimination. The hardest for apraxic clients to differentiate.

8.5.4 Advanced Bilabial Sequencing (Jaw-Lip Dissociation)

A critical component of Phase 2 is separating jaw movement from lip movement. In words with bilabials (/m/, /b/, /p/), clients with poor grading will move their *jaw* up and down to help the lips close, rather than keeping the jaw stable and moving only the lips.⁵

The "Pam" Drill:

- **Target:** The word "Pam" (/p/ - /æ/ - /m/).
- **Jaw Mechanics:** The jaw should drop for the vowel /æ/ and *stay relatively low* or mid-low while the lips close for the final /m/.
- **Error:** The jaw closes completely for the /m/, losing the vowel resonance space.
- **Correction:** Use **Block #5**. Have the client bite the block and say "Pam." The lips must stretch around the block to achieve closure for /p/ and /m/ while the jaw stays fixed. This forces the mentalis and orbicularis oris to work independently of the mandible.³⁷

Bilabial Word List for Dissociation:

- **Map, Mom, Pop, Bob, Pam** (Low Vowels).
- **Baby, Puppy, Bubble** (Mid-High Transitions).
- **Problem:** "Bubble" requires the jaw to stay relatively stable while the lips do the work for /b/.

8.6 Troubleshooting Compensatory Patterns

In Phase 2, as demands increase, the neuromuscular system often reverts to primitive compensatory patterns. The clinician must be vigilant in identifying and correcting these "cheats."

8.6.1 The "Slide" (Lateral Shift)

- **Observation:** When opening for a Low Vowel or Block #6, the jaw slides to the left or right.
- **Mechanism:** Asymmetrical weakness in the lateral pterygoids or TMJ capsular tightness.¹
- **Correction:** Return to **Bilateral Bite** exercises. Use two blocks of the same size, one on each side, to force symmetrical depression. Do not advance to speech drills until symmetry is achieved.²⁴

8.6.2 The "Jut" (Anterior Protrusion)

- **Observation:** The jaw pushes forward when biting or speaking.
- **Mechanism:** Recruitment of the external pterygoids to stabilize the condyle, often seen in Class III malocclusion tendencies or low tone.⁷
- **Correction:** Use the **"Head Flexion" Cue**. Have the client tuck their chin slightly toward their chest during the exercise. This mechanical position inhibits protrusion and encourages retraction, seating the condyles properly.³⁸

8.6.3 Lip Assistance (Mentalis Recruitment)

- **Observation:** The chin skin wrinkles (orange peel effect) or the lower lip pushes up to

help close the mouth or elevate the jaw.

- **Mechanism:** Weakness in the masseter/temporalis; the mentalis muscle attempts to "lift" the lower lip and jaw assembly.
- **Correction: Tactile Anchoring.** The clinician places a finger gently but firmly on the client's mentalis (just below the lip) and pulls down slightly during the exercise. This sensory input inhibits the muscle, forcing the jaw elevators to take the load.⁸

8.6.4 Head Movement (The "Head Bob")

- **Observation:** The head nods forward on jaw closing and tips back on jaw opening.
- **Mechanism:** The client is using the neck muscles (sternocleidomastoid/extension) to drive the jaw because the jaw muscles are functionally disconnected or weak.
- **Correction:** Stabilize the head. The clinician may place a hand on the back of the neck or have the client rest their head against a high-backed chair. Cue: "Head stays like a statue, only the jaw moves".³⁹

8.7 Conclusion of Phase 2 Protocols

Phase 2: Jaw Grading (The Finesse) is the pivot point in orofacial rehabilitation. It transitions the client from crude, binary jaw movements to the sophisticated, graded excursions required for intelligible speech and safe rotary chewing. By systematically integrating reflex suppression (Protocol A), mechanical blocking (Protocol B), functional mastication (Protocol C), and targeted phonetic drilling (Protocol D), the clinician restores the integrity of the "Jaw, Teeth, Speech, and Body Connection."

Success in Phase 2 is defined not by strength alone, but by **control**—the ability to hold a mid-range position, to transition smoothly across the vertical plane, and to dissociate the jaw from the chaotic movements of the hands and body. Only once this finesse is achieved, verified by the ability to produce distinct vowels and manage complex boluses without compensation, should therapy progress to Phase 3 (Speed and Stamina). This disciplined adherence to the hierarchy ensures that the rehabilitation is not merely a collection of exercises, but a restructuring of the neurological foundation of speech and feeding.³

Works cited

1. Jaw Patterns | Beckman Oral Motor, accessed December 7, 2025, <https://beckmanoralmotor.com/wp-content/uploads/2024/08/Jaw-Patterns.pdf>
2. Bite Tube Set (TalkTools) — Definition, Uses & FAQs, accessed December 7, 2025, <https://blog.talktools.com/glossary/bite-tube-set/>
3. TEST – 3-Part Treatment Plan for OPT (Oral Placement Therapy) - TalkTools Training, accessed December 7, 2025, <https://education.talktools.com/quizzes/test-3-part-treatment-plan-for-opt-oral-placement-therapy/>

4. The Role of the Jaw for Feeding and Speech - PediaStaff, accessed December 7, 2025,
<https://www.pediastaff.com/blog/slp/the-role-of-the-jaw-for-feeding-and-speech-18322>
5. Speech Sound Disorders in Children: An Articulatory Phonology Perspective - PMC, accessed December 7, 2025,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC6997346/>
6. Muscle Based Articulation Assessment Instructions | PDF | Vowel | Human Voice - Scribd, accessed December 7, 2025,
<https://www.scribd.com/document/732832657/Muscle-Based-Articulation-Assessment-Instructions>
7. TalkTools® Jaw Grading Bite Blocks™ (Red), accessed December 7, 2025,
<https://talktools.com/products/jaw-grading-bite-blocks>
8. Improving Jaw Strength & Stability - ARK Therapeutic, accessed December 7, 2025,
<https://www.arktherapeutic.com/blog/improving-jaw-strength-stability/>
9. Jaw Stability and Speech: Why It Matters More Than You Think, accessed December 7, 2025,
<https://www.simonsaysspeech.com/myomatters/jawstabilityblog>
10. OROFACIAL FUNCTION IN CHILDREN WITH SPEECH SOUND DISORDERS - KI Open Archive, accessed December 7, 2025,
<https://openarchive.ki.se/ndownloader/files/48952816>
11. Vowels Distortions in Childhood Apraxia of Speech, accessed December 7, 2025,
<https://www.bondarspeech.com/blog/vowels-distortions-in-childhood-apraxia-of-speech>
12. Primitive Reflexes: The Hidden Key to Speech and Language Development, accessed December 7, 2025,
<https://www.rosebudspeech.com/post/primitive-reflexes-the-hidden-key-to-speech-and-language-development>
13. Reflexes That Impact Speech and Feeding Development - Unity Therapy Solutions, accessed December 7, 2025,
<https://www.unitytherapysolutions.com/blog/blog-post-title-three-gdz8h-s7fan>
14. Retained Neonatal Reflexes: Home, accessed December 7, 2025,
<https://rnrs.com.au/>
15. Impact of the orofacial area reflexes on infant's speech development, accessed December 7, 2025,
https://www.umb.edu.pl/photo/pliki/progress-file/phs/phs_2014_1/188-194_kondraciuk.pdf
16. Go with the Flow: How the Babkin Reflex helps your Baby fill their Belly, accessed December 7, 2025,
<https://www.fledglingsflight.com/post/babkin-reflex>
17. Primitive Reflex Integration: Free Expert-Backed Guide - Soundsory, accessed December 7, 2025,
<https://soundsory.com/primitive-reflex-integration/>
18. Why should my child's SLP care about reflex integration? - Unity Therapy Solutions, accessed December 7, 2025,
<https://www.unitytherapysolutions.com/blog/why-should-my-slp-care-about>
19. Primitive Reflexes, Neurodevelopmental Therapy, Primitive Reflex Integration,

- Primitive Reflexes - Brain & Body Boost, accessed December 7, 2025,
<https://brainbodyboost.co.uk/primitive-reflexes?blogcategory=Work>
20. Babkin Reflex Integration Exercise - YouTube, accessed December 7, 2025,
<https://www.youtube.com/watch?v=pNHWGj6Gox0>
 21. Oral Motor Therapy with Babkin Reflex - Reflex Integration Therapy - YouTube, accessed December 7, 2025, <https://www.youtube.com/watch?v=gSMtxzmCvKk>
 22. Free Guide I Primitive Reflex Integration Exercises for Children - Soundsory, accessed December 7, 2025,
<https://soundsory.com/primitive-reflex-integration-exercises/>
 23. Primitive Reflex Integration - The Autism Community in Action, accessed December 7, 2025,
<https://tacanow.org/family-resources/primitive-reflex-integration/>
 24. Jaw Grading Oral Motor Exercises with ARK's Bite Blocks - ARK ..., accessed December 7, 2025,
<https://www.arktherapeutic.com/blog/jaw-grading-oral-motor-exercises-with-arks-bite-blocks/>
 25. Speech Blocks Instructions | PDF - Scribd, accessed December 7, 2025,
<https://www.scribd.com/document/763656600/Speech-Blocks-Instructions>
 26. Jaw Grading & Stability - ARK Therapeutic, accessed December 7, 2025,
<https://www.arktherapeutic.com/blog/jaw-grading-stability/>
 27. Oral Muscle Tone In Speech and Feeding - Playstreet, accessed December 7, 2025,
<https://www.playstreet.in/2021/01/18/oral-muscle-tone-in-speech-and-feeding/>
 28. Stabilizing the Jaw: Best Method for Articulation Therapy - Marshalla Speech & Language, accessed December 7, 2025,
<https://pammarshalla.com/stabilizing-the-jaw-best-method-for-articulation-therapy/>
 29. Chewing Hierarchy p. 1.jpg-1.jpeg - Alphabet Soup, accessed December 7, 2025,
<https://alphabetsoup-j5mb.squarespace.com/s/Chewing-Hierarchy-1-and-2-PDF.pdf>
 30. Pre-Feeding Chewing Hierarchy Level #3: Present one yellow Chewy Tube - Alphabet Soup, accessed December 7, 2025,
<https://alphabetsoup-j5mb.squarespace.com/s/PreFeeding-Hierarchy-3-and-4-PDF.pdf>
 31. Four step chewing hierarchy Level 4 - YouTube, accessed December 7, 2025,
<https://www.youtube.com/watch?v=ql9K12Bx8Qw>
 32. (PDF) Diet-Shaping, Merkel-Walsh and Overland Diet-Shaping: A Useful Technique for Children on the Autism Spectrum Robyn Merkel-Walsh MA, CCC-SLP Lori L. Overland MS, CCC-SLP/NDT-C - ResearchGate, accessed December 7, 2025,
https://www.researchgate.net/publication/338954863_Diet-Shaping_Merkel-Walsh_and_Overland_Diet-Shaping_A_Useful_Technique_for_Children_on_the_Autism_Spectrum_Robyn_Merkel-Walsh_MA_CCC-SLP_Lori_L_Overland_MS_CCC-SLP/NDT-C
 33. Full article: Range of motion (ROM) in the lips and jaw during vowels assessed

with 3D motion analysis in Swedish children with typical speech development and children with speech sound disorders - Taylor & Francis Online, accessed December 7, 2025,

<https://www.tandfonline.com/doi/full/10.1080/14015439.2021.1890207>

34. Oral Placement Therapy To Improve Speech Clarity and Feeding Skills | PDF | Lip - Scribd, accessed December 7, 2025,
<https://www.scribd.com/document/713318766/Oral-Placement-Therapy-to-Improve-Speech-Clarity-and-Feeding-Skills>
35. An Effective Oral Motor Intervention Protocol for Infants and Toddlers with Low Muscle Tone - Diane Bahr, accessed December 7, 2025,
<https://agesandstages.net/pdf/Binder1.pdf>
36. verbalbehavior / Articulation Word Lists - PBworks, accessed December 7, 2025,
<http://verbalbehavior.pbworks.com/Articulation-Word-Lists>
37. Apraxia Word Lists For Adults: 1,300 Speech Therapy Words, accessed December 7, 2025,
<https://theadultspeechtherapyworkbook.com/apraxia-word-lists-for-adults/>
38. Speech Clarity - Moebius Syndrome Foundation, accessed December 7, 2025,
<https://moebiussyndrome.org/wp-content/uploads/2015/07/SaraRJ-Speech-2014.pdf>
39. OSLHA FALL CONFERENCE September 23-24, 2016 Colette Ellis M.Ed., CCC-SLP, BCS-S Speech-Language Pathologist, accessed December 7, 2025,
https://cdn.ymaws.com/www.summitproservices.com/resource/resmgr/2016_osh_a_conference/PDF_Handouts/Oral_Placement_Therapy_Pract.pdf